

RALEIGH COUNTY AP, WV, USA

WMO: 724120

Lat: 37.784N

Lon: 81.123W

Elev: 766

StdP: 92.45

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03872

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.2	-11.6	-19.9	0.7	-11.1	-17.4	0.9	-9.1	10.3	2.5	9.1	1.5	3.2	310	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	29.3	20.8	28.2	20.4	27.2	20.0	22.5	26.8	21.8	25.9	21.2	25.3	3.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.2	17.4	24.6	20.5	16.6	23.9	19.9	16.0	23.3	70.0	27.0	67.3	26.0	64.9	25.1	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.6	6.9	-18.3	31.1	3.8	1.6	-21.0	32.3	-23.2	33.2	-25.4	34.1	-28.1	35.3		
(5)			-18.7	23.8	3.8	1.0	-21.4	24.6	-23.7	25.2	-25.8	25.7	-28.6	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	-0.3	1.6	5.7	11.9	16.0	19.9	21.5	21.1	17.9	12.2	6.4	2.1	(6)
(7)		DBStd	9.15	6.95	6.32	6.23	5.39	4.30	2.80	2.41	2.39	3.61	4.85	5.48	5.97	(7)
(8)		HDD10.0	1201	324	243	163	43	7	0	0	0	1	34	133	252	(8)
(9)		HDD18.3	2867	576	469	393	201	95	17	5	6	50	196	357	502	(9)
(10)		CDD10.0	1707	7	7	29	98	194	298	357	343	238	102	26	8	(10)
(11)		CDD18.3	332	0	0	1	6	24	65	103	91	37	5	0	0	(11)
(12)		CDH23.3	1918	0	0	4	49	153	359	592	502	230	28	1	0	(12)
(13)		CDH26.7	329	0	0	0	3	17	53	116	91	46	4	0	0	(13)
(14)	Wind	WSAvg	3.0	3.8	3.7	3.7	3.6	3.0	2.5	2.2	2.1	2.4	2.8	3.2	3.4	(14)
(15)	Precipitation	PrecAvg	1070	78	72	94	90	110	102	123	90	86	69	75	81	(15)
(16)		PrecMax	1452	162	177	233	194	194	205	245	204	253	150	170	163	(16)
(17)		PrecMin	829	14	18	37	7	26	32	42	27	12	4	17	14	(17)
(18)		PrecStd	149	35	33	44	35	47	43	49	35	51	35	35	35	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	19.2	23.5	27.0	28.5	29.7	30.9	30.2	30.1	26.0	21.7	18.0	(19)
(20)			MCWB	12.4	12.7	14.1	15.1	18.5	20.6	22.1	21.2	19.1	18.1	14.3	13.2	(20)
(21)		2%	DB	15.0	15.8	20.3	24.8	26.6	28.1	29.2	28.8	28.0	23.5	18.8	15.5	(21)
(22)			MCWB	11.8	10.7	12.4	14.6	17.9	20.5	21.5	21.1	19.1	16.4	12.4	12.0	(22)
(23)		5%	DB	12.6	13.5	17.6	22.8	25.1	26.9	27.9	27.6	26.1	21.7	16.8	13.4	(23)
(24)			MCWB	9.8	9.4	11.1	13.6	17.3	20.0	21.0	20.6	19.0	15.2	11.8	10.5	(24)
(25)		10%	DB	10.0	11.1	15.0	20.6	23.4	25.5	26.7	26.3	24.1	19.8	14.8	11.4	(25)
(26)			MCWB	7.1	7.5	9.7	12.9	16.7	19.4	20.6	20.2	17.9	14.8	10.9	8.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.0	14.3	15.1	17.4	20.6	22.7	23.7	23.2	21.4	20.0	16.0	14.5	(27)
(28)			MCD B	16.2	17.3	21.6	23.7	26.1	27.1	28.3	27.6	26.4	23.5	19.7	16.7	(28)
(29)		2%	WB	12.2	12.2	13.6	16.2	19.5	21.6	22.7	22.4	20.6	18.3	14.4	12.8	(29)
(30)			MCD B	14.6	14.7	18.4	21.9	24.3	26.1	27.1	26.6	25.2	21.9	17.2	14.8	(30)
(31)		5%	WB	10.2	10.1	12.4	15.1	18.6	20.8	22.0	21.6	19.9	16.9	13.0	11.1	(31)
(32)			MCD B	12.3	12.6	16.2	20.2	23.0	25.3	26.0	25.6	23.8	19.9	15.5	13.1	(32)
(33)		10%	WB	7.6	7.7	10.8	14.0	17.6	20.2	21.4	20.9	19.1	15.6	11.4	8.9	(33)
(34)			MCD B	9.5	10.4	14.0	18.7	21.7	24.2	25.2	24.8	22.6	18.6	14.1	10.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.5	9.0	10.0	11.2	10.3	9.6	9.2	9.4	9.9	10.2	9.5	8.1	(35)
(36)			MCD BR	10.1	12.3	12.9	13.4	12.1	11.1	10.7	10.8	12.1	12.1	11.5	10.2	(36)
(37)			MCWBR	7.2	8.6	7.3	6.4	5.4	5.0	4.5	4.7	5.1	6.1	7.1	7.6	(37)
(38)		5% WB	MCD BR	9.1	10.8	10.7	11.0	10.1	9.6	9.0	9.1	9.5	9.5	9.6	9.6	(38)
(39)			MCWBR	7.2	8.8	7.0	5.9	5.1	4.7	4.2	4.3	4.5	5.6	7.3	8.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.290	0.301	0.324	0.362	0.405	0.423	0.455	0.460	0.361	0.335	0.304	0.285	(40)	
(41)		taud	2.516	2.474	2.467	2.378	2.248	2.229	2.169	2.128	2.409	2.482	2.547	2.557	(41)	
(42)		Ebn at Noon	904	935	942	917	878	858	828	813	888	882	880	888	(42)	
(43)		Edh at Noon	77	92	102	118	137	140	148	150	107	90	75	70	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.95	2.63	3.73	4.88	5.45	6.00	5.63	5.25	4.42	3.31	2.41	1.72	(44)	
(45)		RadStd	0.20	0.35	0.28	0.38	0.44	0.39	0.33	0.27	0.43	0.41	0.28	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.50	N/A	N/A	N/A	N/A	N/A	N/A	-147	+150	+66
(47) Regional (30 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137	+71

Nomenclature: See separate page